

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	139782	46.024978
1	[Git, Change, Create]	45409	14.951483
2	[Git, Change, Delete]	22797	7.506198
3	[Java, ByteCode, Member, Method]	13763	4.531640
4	[Java, ByteCode, Parameter]	13447	4.427594
5	[Git, Commit]	11584	3.814177
6	[File, Git]	8331	2.743086
7	[Java, ByteCode, ParameterizedType, Bound]	7316	2.408885
8	[Java, ByteCode, Bound]	7304	2.404934
9	[Git, Change, Rename]	7018	2.310765
10	[Java, ByteCode, Member, Field]	3701	1.218601
11	[Java, Value, ByteCode, Annotation]	2955	0.972971
12	[Java, ByteCode, Bound, WildcardType]	2950	0.971325
13	[Java, ByteCode, Member, Constructor, Method]	2168	0.713841
14	[Xml, Element]	2162	0.711866
15	[Xml, Text]	1450	0.477431
16	[Java, ByteCode, Bound, TypeVariable]	1112	0.366140
17	[Java, ByteCode, Member, Method, Lambda]	986	0.324653
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	903	0.297324
19	[Type, File, Java, Class, ByteCode]	868	0.285800
20	[Json, Key]	739	0.243325
21	[Value, Json, Scalar]	723	0.238057
22	[Java, Value, ByteCode, Primitive]	685	0.225545
23	[Type, File, Java, ByteCode, JavaType]	659	0.216984
24	[Java, ByteCode, Member, Method, GenericDeclar...	579	0.190643
25	[Type, File, Java, ByteCode, ExternalType]	419	0.137961
26	[Git, Change, Copy]	298	0.098120
27	[Author, Git, Person]	297	0.097791
28	[Value, Array]	293	0.096474
29	[Committer, Git, Person]	242	0.079682

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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Nodes per label combination (more than 0.5% overall)

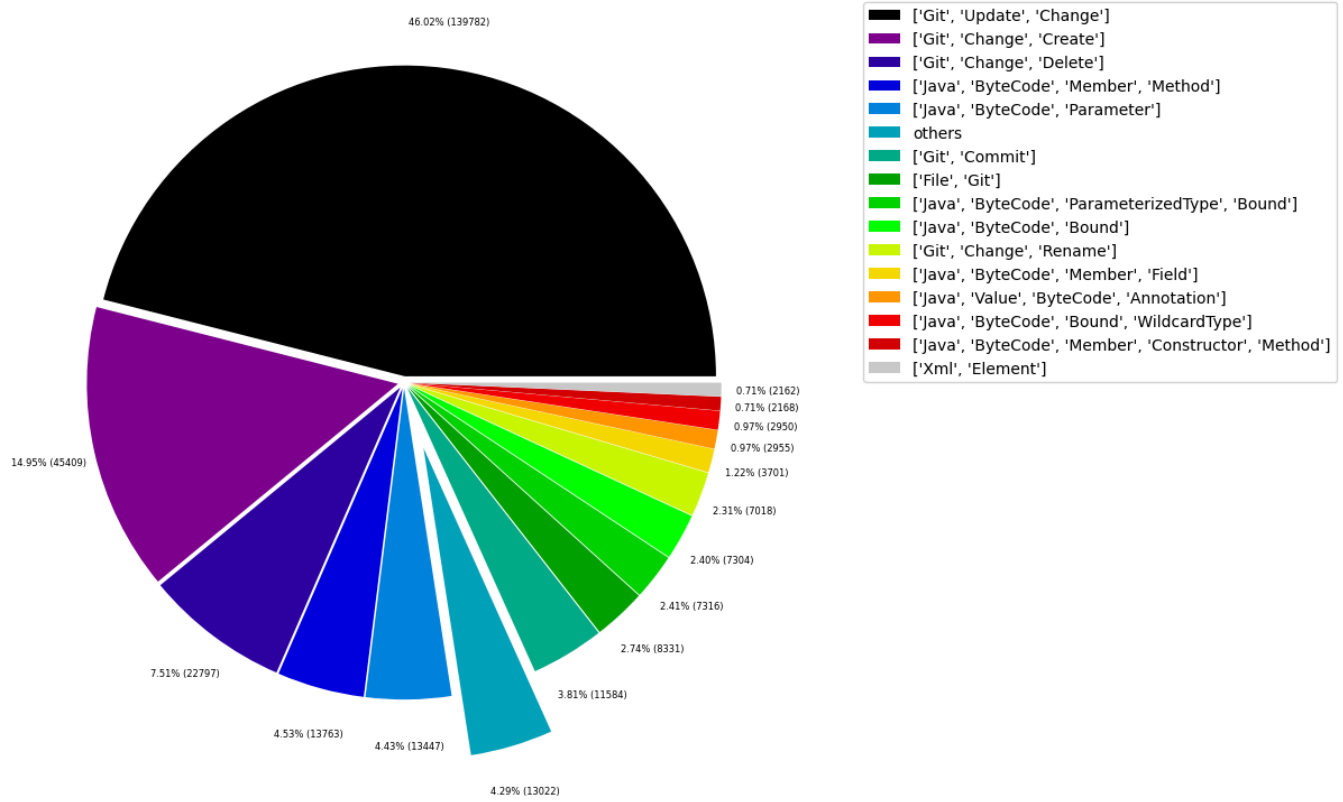


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000329
1	[Git, Branch]	1	0.000329
2	[Repository, File, Git]	1	0.000329
3	[File, Json]	2	0.000659
4	[File]	3	0.000988
5	[Java, ByteCode, Member, Constructor, Method, ...]	4	0.001317
6	[Maven, Exclusion]	5	0.001646
7	[Value, Json, Array]	6	0.001976
8	[File, Maven, Xml, Pom, Document]	9	0.002963
9	[Type, File, Java, ByteCode, Void]	9	0.002963
10	[Java, ManifestSection]	9	0.002963
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002963
12	[File, Java, Manifest]	9	0.002963
13	[File, Java, ServiceLoader]	11	0.003622
14	[File, Java, Properties]	12	0.003951
15	[Maven, ExecutionGoal]	16	0.005268
16	[Maven, PluginExecution]	16	0.005268
17	[Xml, Attribute]	18	0.005927
18	[JQAssistant, Rule, Concept]	19	0.006256
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006915
20	[Maven, Plugin]	21	0.006915
21	[Maven, Configuration]	21	0.006915
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007244
23	[Type, File, Java, ByteCode, Enum]	30	0.009878
24	[Type, File, Java, ByteCode, PrimitiveType]	31	0.010207
25	[Xml, Namespace]	36	0.011853
26	[Type, File, Java, ByteCode, Annotation]	44	0.014488
27	[File, Directory]	45	0.014817
28	[Type, File, Java, Class, ByteCode, Throwable]	56	0.018439
29	[Type, File, Java, ByteCode, Interface, Generi...	85	0.027987

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

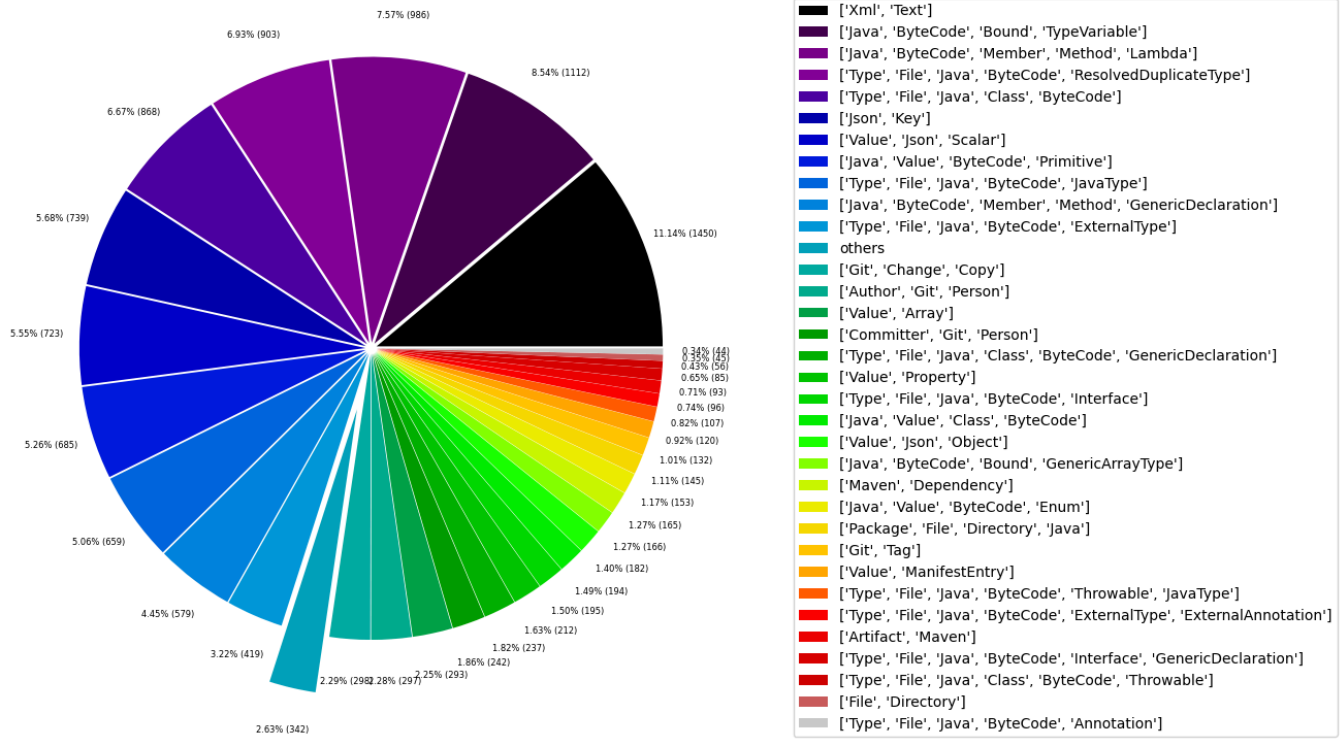


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

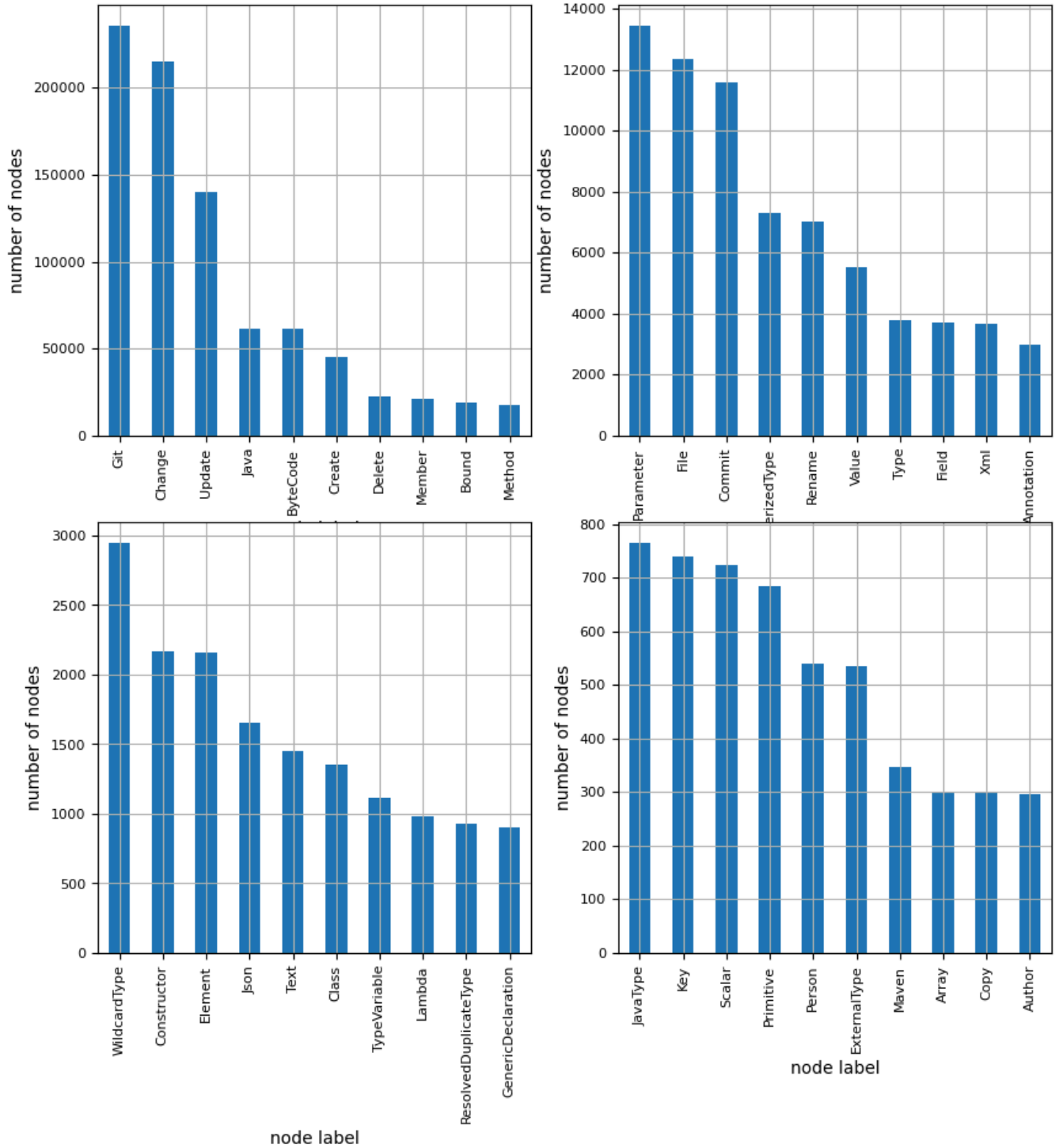
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	235892	77.670402
1	Change	215304	70.891544
2	Update	139782	46.024978
3	Java	61460	20.236476
4	ByteCode	61265	20.172270
5	Create	45409	14.951483
6	Delete	22797	7.506198
7	Member	21201	6.980695
8	Bound	18848	6.205941
9	Method	17500	5.762095
10	Parameter	13447	4.427594
11	File	12359	4.069356
12	Commit	11584	3.814177
13	ParameterizedType	7316	2.408885
14	Rename	7018	2.310765
15	Value	5523	1.818517
16	Type	3782	1.245271
17	Field	3701	1.218601
18	Xml	3675	1.210040
19	Annotation	2999	0.987458
20	WildcardType	2950	0.971325
21	Constructor	2172	0.715158
22	Element	2162	0.711866
23	Json	1652	0.543942
24	Text	1450	0.477431
25	Class	1355	0.446151
26	TypeVariable	1112	0.366140
27	Lambda	986	0.324653
28	ResolvedDuplicateType	925	0.304568
29	GenericDeclaration	905	0.297983
30	JavaType	766	0.252215
31	Key	739	0.243325
32	Scalar	723	0.238057
33	Primitive	685	0.225545
34	Person	539	0.177473
35	ExternalType	536	0.176485
36	Maven	346	0.113925
37	Array	299	0.098450
38	Copy	298	0.098120
39	Author	297	0.097791

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 968016

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	215304	22.241781
1	MODIFIES	215304	22.241781
2	UPDATES	139782	14.440051
3	CREATES	52725	5.446707
4	INVOKES	37783	3.903138
5	DELETES	29815	3.080011
6	CHANGED_TOGETHER_WITH	26193	2.705844
7	COMMITTED	23168	2.393349
8	DEPENDS_ON	22889	2.364527
9	OF_TYPE	22204	2.293764
10	DECLARES	21666	2.238186
11	OF_RAW_TYPE	17460	1.803689
12	HAS	14602	1.508446
13	HAS_PARENT	14170	1.463819
14	RETURNS	13073	1.350494
15	HAS_COMMIT	11584	1.196674
16	READS	9677	0.999674
17	HAS_ACTUAL_TYPE_ARGUMENT	8445	0.872403
18	HAS_FILE	8331	0.860626
19	RENAMES	7018	0.724988
20	OF_GENERIC_TYPE	6036	0.623543
21	RESOLVES_TO	4303	0.444517
22	SIMILAR	4130	0.426646
23	WRITES	4042	0.417555
24	CONTAINS	4005	0.413733
25	HAS_NEW_NAME	3662	0.378300
26	RETURNS_GENERIC	3616	0.373548
27	ANNOTATED_BY	2943	0.304024
28	REQUIRES	2267	0.234190
29	HAS_FIRST_CHILD	2162	0.223343

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

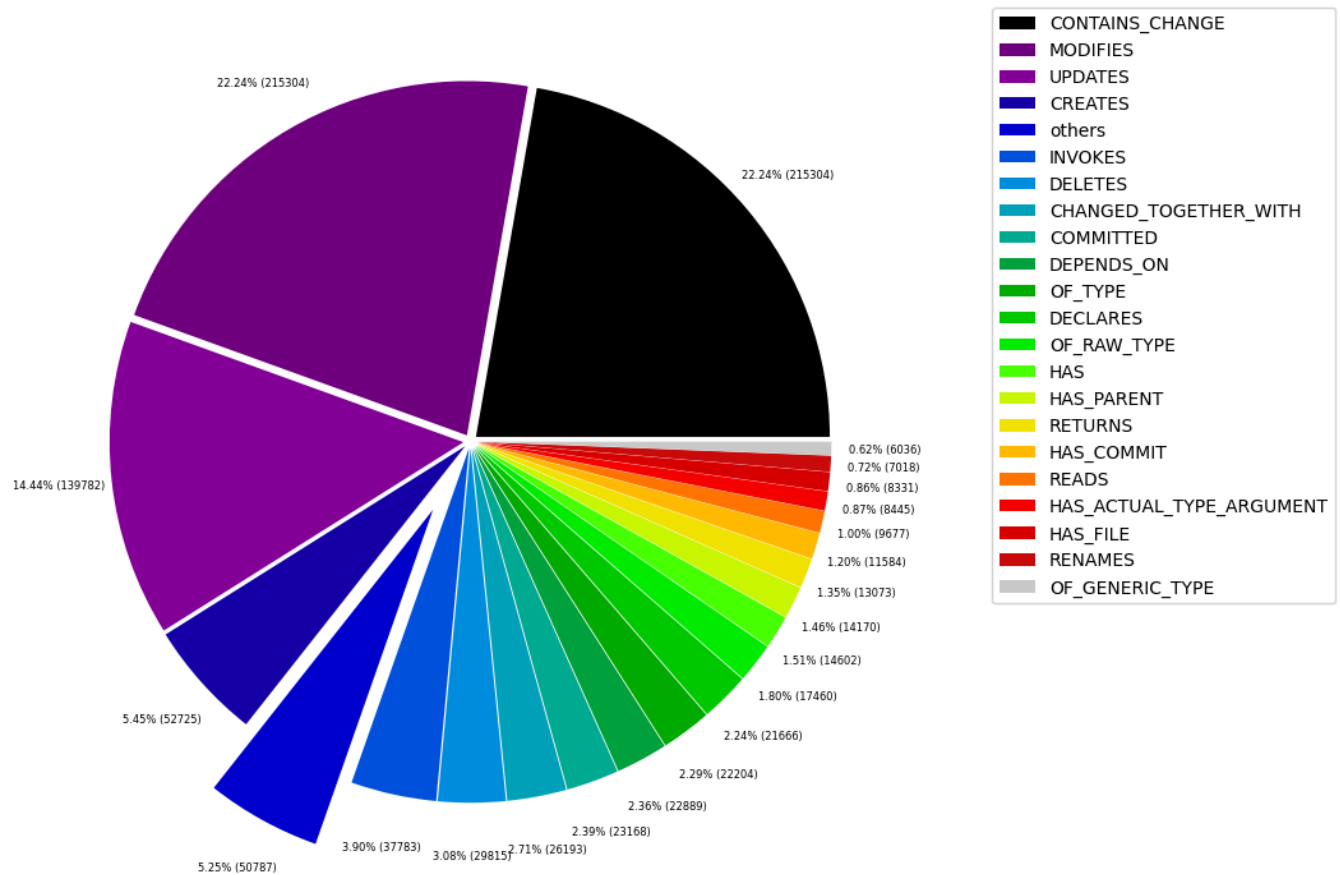


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

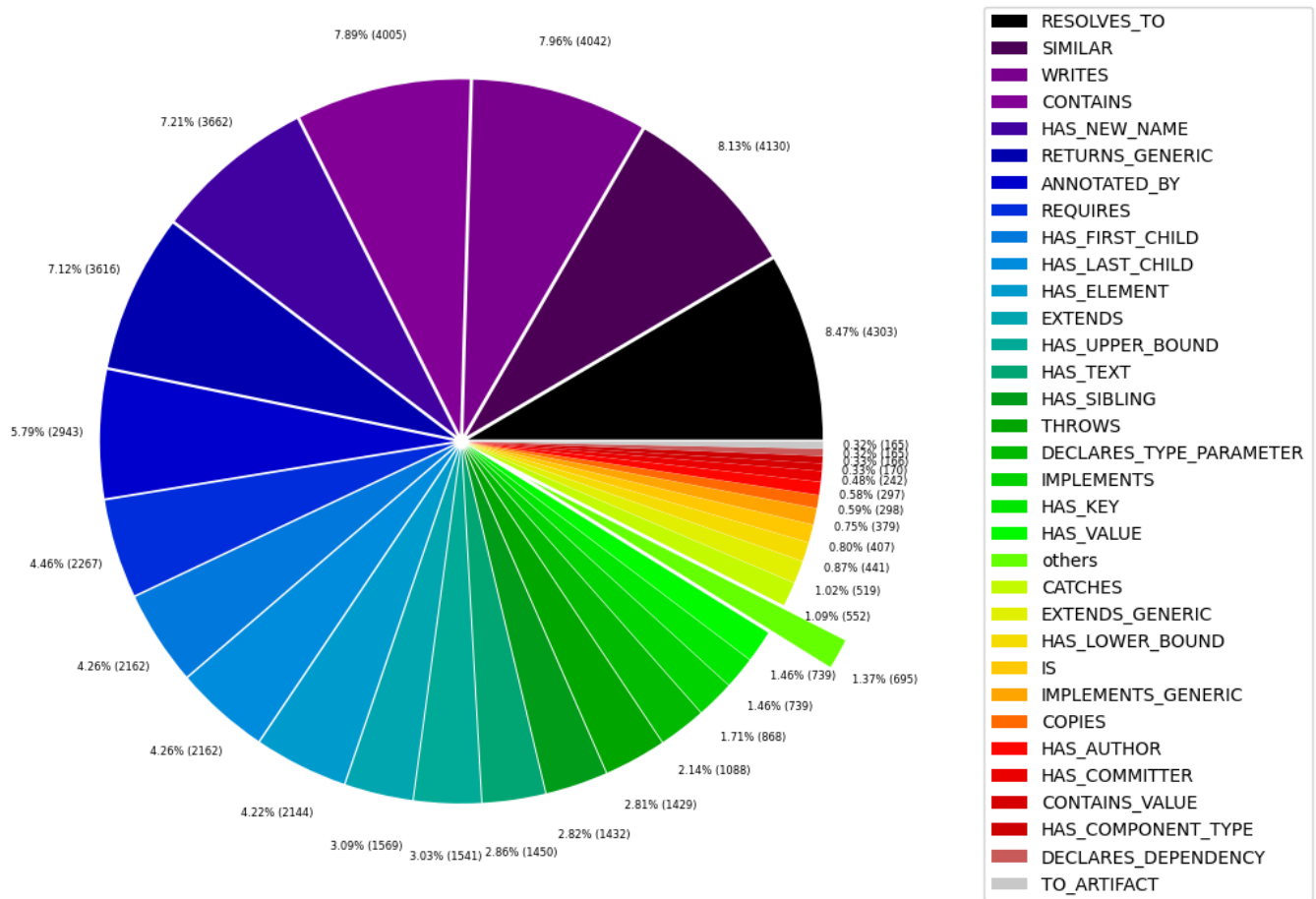
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000103
1	HAS_BRANCH	1	0.000103
2	HAS_HEAD	2	0.000207
3	THROWS_GENERIC	5	0.000517
4	EXCLUDES	5	0.000517
5	DESCRIBES	9	0.000930
6	HAS_ROOT_ELEMENT	12	0.001240
7	HAS_GOAL	16	0.001653
8	HAS_EXECUTION	16	0.001653
9	OF_NAMESPACE	18	0.001859
10	HAS_ATTRIBUTE	18	0.001859
11	INCLUDES_CONCEPT	19	0.001963
12	USES_PLUGIN	21	0.002169
13	IS_ARTIFACT	21	0.002169
14	HAS_CONFIGURATION	21	0.002169
15	REQUIRES_TYPE_PARAMETER	24	0.002479
16	REQUIRES_CONCEPT	28	0.002893
17	DECLARES_NAMESPACE	36	0.003719
18	HAS_DEFAULT	39	0.004029
19	COPY_OF	119	0.012293
20	HAS_TAG	132	0.013636
21	ON_COMMIT	132	0.013636
22	DECLARES_DEPENDENCY	165	0.017045
23	TO_ARTIFACT	165	0.017045
24	HAS_COMPONENT_TYPE	166	0.017148
25	CONTAINS_VALUE	170	0.017562
26	HAS_COMMITTER	242	0.025000
27	HAS_AUTHOR	297	0.030681
28	COPIES	298	0.030785
29	IMPLEMENTS_GENERIC	379	0.039152

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	139782	11584	
1	[Git, Update, Change]	MODIFIES	[File, Git]	139782	139782	
2	[Git, Update, Change]	UPDATES	[File, Git]	139782	139782	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	45409	11584	
4	[Git, Change, Create]	MODIFIES	[File, Git]	45409	45409	
5	[Git, Change, Create]	CREATES	[File, Git]	45409	45409	
6	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	23045	13662	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22797	11584	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22797	22797	
9	[Git, Change, Delete]	DELETES	[File, Git]	22797	22797	
10	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	20059	8331	
11	[Git, Commit]	HAS_PARENT	[Git, Commit]	14161	11584	
12	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11584	1	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11584	242	
14	[Author, Git, Person]	COMMITTED	[Git, Commit]	11584	297	
15	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8652	13662	
16	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8592	13662	
17	[Repository, File, Git]	HAS_FILE	[File, Git]	8331	1	
18	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11584	
19	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
22	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6235	13447	
24	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8331	
25	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3488	7304	
26	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3144	7316	
27	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7316	
28	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2709	13447	
29	[Java, ByteCode, Member, Constructor, Method]	WRITES	[Java, ByteCode, Member, Field]	2577	2168	

Graph Density

```
total_number_of_nodes (vertices): 303709
total_number_of_relationships (edges): 968016
-> total directed graph density: 1.0494666496924574e-05
-> total directed graph density in percent: 0.0010494666496924574
```